FN585 Financial Modelling & Dealing

Week 4

Shahab Gharaati

University of Brighton

Assumptions for OLS in Distributed Lag (DL) Models

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \cdots + u_t$$

- **[C1]** The model is linear in the parameters and correctly specified
- **[C2]** The time series (Y_t, X_t) are weakly stationary and (Y_t, X_t) and (Y_{t-j}, X_{t-j}) become independent as j gets large (weak dependence)
- **[C3]** There is no exact linear relationship among regressors (no perfect multicollinearity)
- **[C4] Exogeneity:** $E(u_t|X_t, X_{t-1}, \dots) = 0$ for all t: conditioning set is current and past values of X
- **[C4-Strict] Strict Exogeneity:** $E(u_t|\dots, X_{t+1}, X_t, X_{t-1}, \dots) = 0$ for all t: conditional on past, current, and future X

Assumptions for OLS in distributed lag (ADL) models

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \cdots + u_t$$

Under C1-C4-strict, the OLS of β_1 is unbiased. However, strict **exogeneity** is a strong condition that needs to be relaxed in many economic applications.

Under C1-C4, OLS is consistent, although not unbiased. For T large enough, we are happy with consistency.

Autoregressive distributed lag Models

Addittional Predictors

- Autoregressive forecasts of Y use the immediate past values of Y itself; often, economic theory suggests another variable X may help forecast Y as well.
- Autoregressive Distributed Lag models adds additional predictors to the forecasting model
  - "Autoregressive" due to lags of dependent variable
- Also includes a lag distribution of additional predictor X
- A parsimonious (small no. of parameters) way of incorporating dynamic relationship across time: alleviates problem of multicollinearity when including too many lags of X_t

Autoregressive distributed lag Models

- Consider two time series Y_t and X_t, and consider:

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots + \beta_p Y_{t-p} +$$

$$\delta_1 X_{t-1} + \delta_2 X_{t-2} + \cdots + \delta_q X_{t-q} + u_t,$$

  where $\beta_0, \beta_1, \ldots, \beta_p, \delta_1, \delta_2, \ldots, \delta_q$ are coefficients, and u_t is the error term with the property:

$$E(u_t | Y_{t-1}, Y_{t-2}, \ldots, X_{t-1}, X_{t-2}, \ldots) = 0$$

- This model is an ADL(p, q)-model
- The ADL(p, q)-model is a multiple regression model

GDP growth and the term spread

```r
usmac_qt <- read_xlsx("us_macro_quarterly.xlsx",sheet = 1,
                      col_types = c("text", rep("numeric", 9)))
```

```
## New names:
## * `` -> `...1`
```

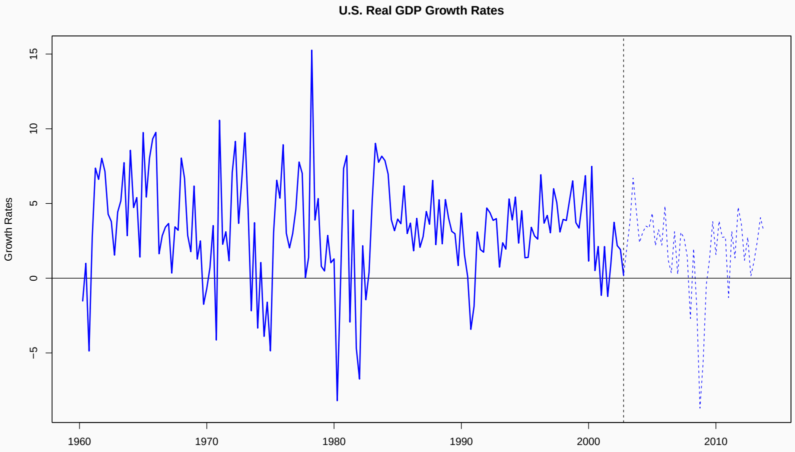
```r
# Fix format of date
usmac_qt$...1 <- as.yearqtr(usmac_qt$...1, format = "%Y:0%q")
# Relabel column names in dataframe
colnames(usmac_qt) <- c("Date", "GDPC96", "JAPAN_IP", "PCECTPI",
"GS10", "GS1", "TB3MS", "UNRATE", "EXUSUK",
"CPIAUCSL")
# GDP series as xts object
GDP <- xts(usmac_qt$GDPC96, usmac_qt$Date)["1960::2013"]
# GDP growth series as xts object
GDPGrowth <- xts(400 * log(GDP/lag(GDP)))
# 3-months Tbill
ishort <- xts(usmac_qt$TB3MS, usmac_qt$Date)["1960::2013"]
# 10 years government bond
ilong <- xts(usmac_qt$GS10, usmac_qt$Date)["1960::2013"]
#Term spread
term.spread <- ilong - ishort
```

GDP growth and the term spread

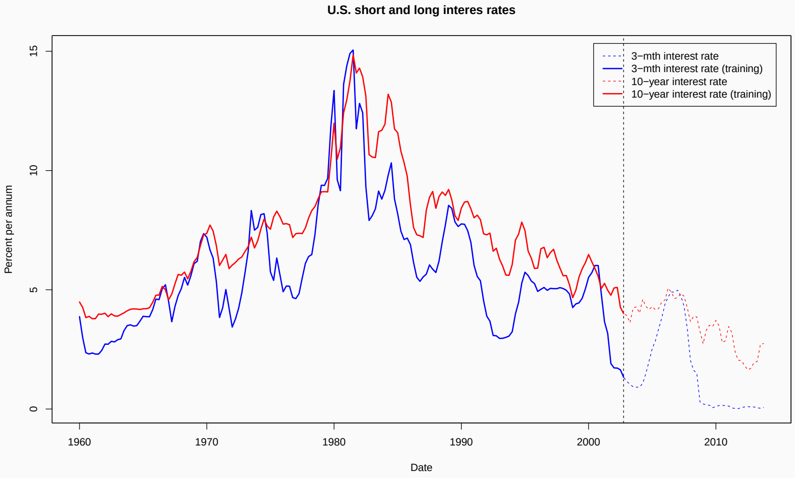
```r
# Training data
GDPGrowth.train <- na.omit(GDPGrowth["1960::2002"])
ishort.train <- na.omit(ishort["1960::2002"])
ilong.train <- na.omit(ilong["1960::2002"])
term.spread.train <- na.omit(term.spread["1960::2002"])
```

The US GDP growth

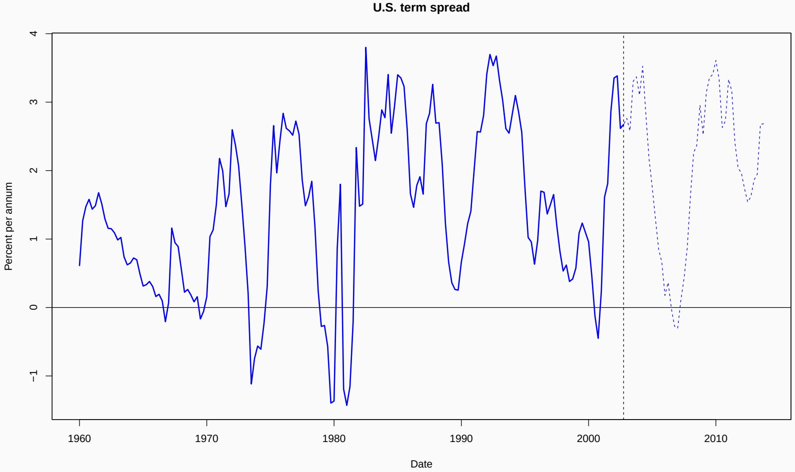
```
# Plot US quarterly GDP time series w/ training data
plot(as.zoo(GDPGrowth), col = "blue", lwd = 1, lty = "dashed",
     ylab = "Growth Rates", xlab = "Date", main = "U.S. Real GDP Growth Rates")
abline(v = 2002.75, lty = "dashed")
abline(h = 0, lwd = 1)
lines(as.zoo(GDPGrowth.train), lwd = 2, col = "blue")
```



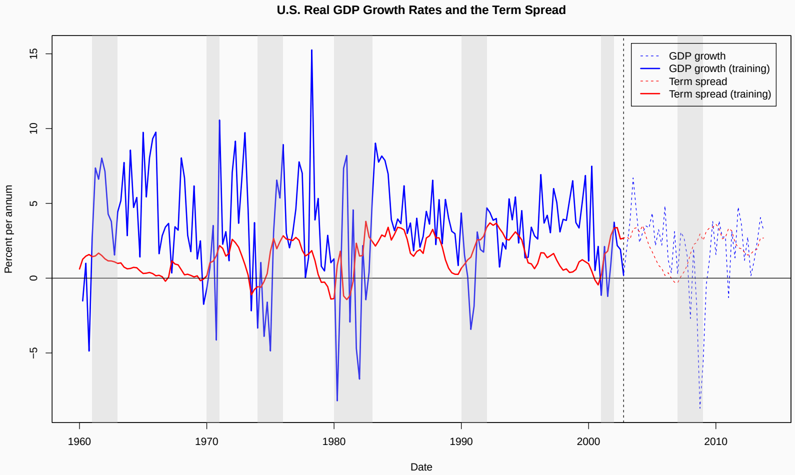
The US short-term and long-term interest rates



The US term spread



The US GDP growth and term spread



ADL(4,4) model for US GDP growth

- Consider the following ADL(4,4)-model for US GDP growth:

$$GDPGR_t = \beta_0 + \sum_{j=1}^{4} \beta_j GDPGR_{t-j} + \sum_{k=1}^{4} \delta_k TSPR_{t-k} + u_t$$

- Estimate using $\{(GDPGR_t, TSPR_t); t = 1, 2, \ldots, T\}$, make a 1-step ahead forecast:

$$\widehat{GDPGR}_{T+1|T} = \hat{\beta}_0 + \sum_{j=0}^{3} \hat{\beta}_j GDPGR_{T-j} + \sum_{k=0}^{3} \hat{\delta}_k TSPR_{T-k}$$

ADL forecasting of GDP growth

```r
# Estimating ADL(4,4)-model for US GDP growth
library(dynlm) # Load dynlm to estimate higher-order autoregressions
library(sandwich)
library(stargazer)
ADL44.dynlm <- dynlm(ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)) + L(ts(GDPGrowth.train), 2) +
                             L(ts(GDPGrowth.train), 3) + L(ts(GDPGrowth.train), 4) +
                             L(ts(term.spread.train)) + L(ts(term.spread.train), 2) +
                             L(ts(term.spread.train), 3) + L(ts(term.spread.train), 4))


# Output w/ robust standard errors
ADL44.Robdynlm<-coeftest(ADL44.dynlm, vcov. = vcovHAC)
```

ADL forecasting of GDP growth

```
stargazer(ADL44.dynlm,ADL44.Robdynlm, type="text")
```

```
##
## ==============================================================
##                                     Dependent variable:
##                             ----------------------------------
##                              ts(GDPGrowth.train)
##                                   dynamic        coefficient
##                                    linear           test
##                                     (1)              (2)
## --------------------------------------------------------------
## L(ts(GDPGrowth.train))            0.218***         0.218**
##                                   (0.080)          (0.092)
##
## L(ts(GDPGrowth.train), 2)          0.136*           0.136
##                                   (0.080)          (0.091)
##
## L(ts(GDPGrowth.train), 3)          -0.017           -0.017
##                                   (0.081)          (0.067)
##
## L(ts(GDPGrowth.train), 4)          0.082            0.082
##                                   (0.079)          (0.085)
##
## L(ts(term.spread.train))          1.341***        1.341***
##                                   (0.446)          (0.502)
##
## L(ts(term.spread.train), 2)        -0.976           -0.976
##                                   (0.652)          (0.773)
##
```

ADL forecasting of GDP growth

	$GDPGrowth_t$	
	Model 1	*Model 2*
$GDPGrowth_{t-1}$	0.218^{***}	0.218^{**}
	(0.080)	(0.092)
$GDPGrowth_{t-2}$	0.136^{*}	0.136
	(0.080)	(0.091)
$GDPGrowth_{t-3}$	-0.017	-0.017
	(0.081)	(0.067)
$GDPGrowth_{t-4}$	0.082	0.082
	(0.079)	(0.085)
$term.spread_{t-1}$	1.341^{***}	1.341^{***}
	(0.446)	(0.502)
$term.spread_{t-2}$	-0.976	-0.976
	(0.652)	(0.773)
$term.spread_{t-3}$	0.666	0.666
	(0.652)	(0.729)
$term.spread_{t-4}$	-0.379	-0.379
	(0.462)	(0.435)
Constant	1.087^{**}	1.087^{*}
	[illegible]	[illegible]

ADL forecasting of GDP growth

1-step ahead ADL(4,4) forecast of US GDP growth

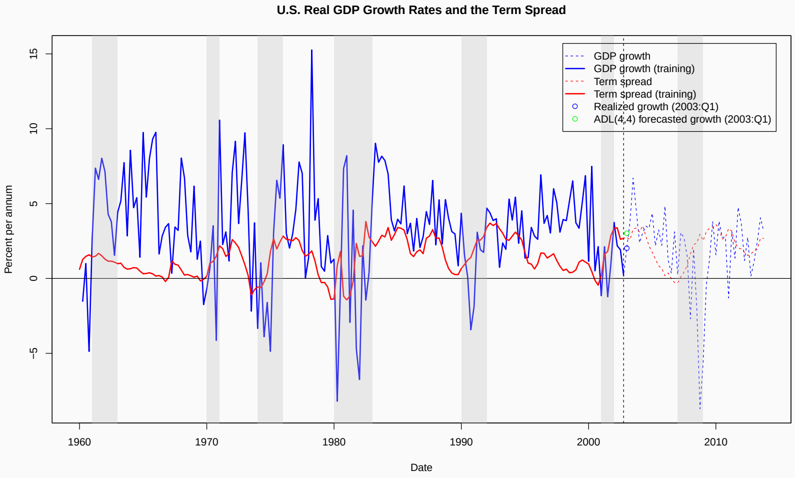
```
# Produce one-step ahead forecast of GDP growth for 2001:Q1
K <-length(GDPGrowth.train) # Length of observed series
fcast.ADL44 <- ADL44.dynlm$coefficients %*% c(1, GDPGrowth.train[K],
                GDPGrowth.train[(K-1)], GDPGrowth.train[(K-2)], GDPGrowth.train[(K-3)],
               term.spread.train[K], term.spread.train[(K-1)], term.spread.train[(K-2)],
               term.spread.train[(K-3)])
fcast.ADL44 # Print to console
```

```
##          [,1]
## [1,] 3.018917
```

```
# Forecast error
fcast.ADL44.error <- as.numeric(GDPGrowth["2003"][1]) - fcast.ADL44
fcast.ADL44.error # Print to console
```

```
##            [,1]
## [1,] -0.9974227
```

ADL forecasting of GDP growth



ADL(2,2)-model for us gdp growth

- Consider the following ADL(2,2)-model for US GDP growth:

$$GDPGR_t = \beta_0 + \sum_{j=1}^{2} \beta_j GDPGR_{t-j} + \sum_{k=1}^{2} \delta_k TSPR_{t-k} + u_t$$

- Estimate using $\{(GDPGR_t, TSPR_t); t = 1, 2, \ldots, T\}$, make a 1-step ahead forecast:

$$\widehat{GDPGR}_{T+1|T} = \hat{\beta}_0 + \sum_{j=0}^{1} \hat{\beta}_j GDPGR_{T-j} + \sum_{k=0}^{1} \hat{\delta}_k TSPR_{T-k}$$

ADL(2,2) model for US GDP growth

```
# Estimating ADL(2,2)-model for US GDP growth
ADL22.dynlm <- dynlm(ts(GDPGrowth.train) ~ L(ts(GDPGrowth.train)
            +L(ts(GDPGrowth.train),2)+ L(ts(term.spread.train))
            +L(ts(term.spread.train),2))
ADL22.Robdynlm<-coeftest(ADL22.dynlm, vcov. = vcovHAC)
```

ADL(2,2) model for US GDP growth

```r
stargazer(ADL22.dynlm, ADL22.Robdynlm, type="text")
```

```
##
## ==============================================================
##                                  Dependent variable:
##                             ----------------------------------
##                              ts(GDPGrowth.train)
##                                    dynamic        coefficient
##                                     linear            test
##                                      (1)              (2)
## --------------------------------------------------------------
## L(ts(GDPGrowth.train))             0.204***         0.204**
##                                    (0.077)          (0.082)
##
## L(ts(GDPGrowth.train), 2)          0.174**          0.174**
##                                    (0.076)          (0.087)
##
## L(ts(term.spread.train))           1.134***         1.134**
##                                    (0.427)          (0.490)
##
## L(ts(term.spread.train), 2)         -0.477           -0.477
##                                    (0.449)          (0.566)
##
## Constant                           1.185***         1.185**
##                                    (0.441)          (0.534)
##
## --------------------------------------------------------------
## Observations                         169
## R2                                  0.182
```

ADL(2,2) model for US GDP growth

	Dependent variable:	
	$GDPGrowth_t$	
	Model 1	Model 2
	(1)	(2)
$GDPGrowth_{t-1}$	0.204***	0.204**
	(0.077)	(0.082)
$GDPGrowth_{t-2}$	0.174**	0.174**
	(0.076)	(0.087)
$term.spread_{t-1}$	1.134***	1.134**
	(0.427)	(0.490)
$term.spread_{t-2}$	−0.477	−0.477
	(0.449)	(0.566)
Constant	1.185***	1.185**
	(0.441)	(0.534)
Observations	169	
R^2	0.177	
Adjusted R^2	0.162	
Residual Std. Error	3.169	
F Statistic	11.810***	
Note:	Model 1 uses non-robust standard errors; Model 2 uses HAC standard errors.	
	*p<0.1; **p<0.05; ***p<0.01	

1-step ahead ADL(2,2) forecast of US GDP growth

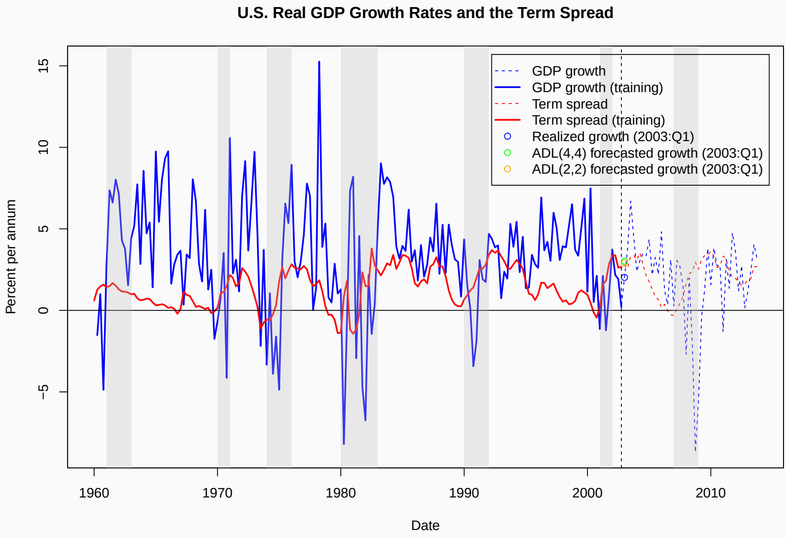
```
# Produce one-step ahead forecast of GDP growth for 2001:Q1
fcast.ADL22 <- ADL22.dynlm$coefficients %*% c(1, GDPGrowth.train[K],
  GDPGrowth.train[K-1],term.spread.train[K],term.spread.train[K-1])
fcast.ADL22 # Print to console
```

```
##          [,1]
## [1,] 2.912231
```

```
# Forecast error
fcast.ADL22.error <- as.numeric(GDPGrowth["2003"][1]) - fcast.ADL22
fcast.ADL22.error # Print to console
```

```
##            [,1]
## [1,] -0.8907367
```

Some examples of economic time series



Motivation for ADL(1,0): partial adjustment, inertia

- Due to inertia, Y_t is a compromise between Y_{t-1} and unobserved $Y_t^* = \gamma_1 + \gamma_2 X_t$, value justified by the current X_t.

$$Y_t = \lambda Y_t^* + (1-\lambda) Y_{t-1}$$

$$\Delta Y_t = Y_t - Y_{t-1} = \lambda (Y_t^* - Y_{t-1})$$

- $Y_t = \gamma_1 \lambda + \gamma_2 \lambda X_t + (1-\lambda) Y_{t-1} + \lambda u_t$ This is ADL(1,0)

Partial adjustment, e.g. Brown's habit formation

- C_t^*: desired level of consumption, W_t: household wage income, NW_t: non-wage income

$$C_t^* = \beta_1 + \beta_2 W_t + \beta_3 NW_t + u_t$$

Due to inertia,

$$C_t - C_{t-1} = \lambda(C_t^* - C_{t-1})$$

- Leading to

$$C_t = \beta_1\lambda + \beta_2\lambda W_t + \beta_3\lambda NW_t + (1-\lambda)C_{t-1} + \lambda u_t$$

Brown (1952) obtained the following fitted model:

$$\hat{C}_t = 0.9 + 0.61 W_t + 0.28 NW_t + 0.22 C_{t-1}$$

Motivation for ADL(1,1): error correction model

- $Y_t^* = \gamma_1 + \gamma_2 X_t$ Error correction:

$$\Delta Y_t = \lambda(Y_t^* - Y_{t-1}) + \delta \Delta X_t + u_t$$

$$= \lambda\gamma_1 + \delta X_t + (\lambda\gamma_2 - \delta)X_{t-1} - \lambda Y_{t-1} + u_t$$

  This is ADL(1,1). λ is speed of adjustment

ADL and equilibrium

- E.g. ADL(1,0)

$$Y_t = \beta_1 + \beta_2 X_t + \beta_3 Y_{t-1} + u_t$$

In equilibrium:

$$Y^e = \frac{\beta_1}{1-\beta_3} + \frac{\beta_2}{1-\beta_3} X^e$$

Caution

- When lagged dependent variables (e.g. Y_{t-1}) appear on RHS, we cannot allow for serial correlation in the error term, as this will lead to inconsistency of OLS! For illustration, consider

$$Y_t = \beta_1 + \beta_2 Y_{t-1} + u_t$$

$$Y_{t-1} = \beta_1 + \beta_2 Y_{t-2} + u_{t-1}$$

  Suppose

$$u_t = \rho u_{t-1} + \epsilon_t$$

  This leads to endogeneity of regressor Y_{t-1}, hence inconsistency of OLS $E(u_t|Y_{t-1}) \neq 0$

Caution

- For OLS to be consistent when lagged dependent variable (e.g. Y_{t-1}) is included in the RHS, we need strict exogeneity (C4-strict) for OLS to be consistent. (You can find reasons and derivation on SW p.625-626 but this won't be covered in this module)

Dynamic causal effects

- The estimated ADL coefficients are not themselves dynamic multipliers.
- But dynamic multipliers can be computed from ADL coefficients.
- E.g. ADL(1,1):

$$Y_t = \beta_1 Y_{t-1} + \delta_1 X_{t-1} + u_t$$

$$= \beta_1(\beta_1 Y_{t-2} + \delta_1 X_{t-2} + u_{t-1}) + \delta_1 X_{t-1} + u_t$$

$$= \delta_1 X_{t-1} + \beta_1 \delta_1 X_{t-2} + \beta_1 u_{t-1} + u_t + \beta_1^2(\beta_1 Y_{t-3} + \delta_1 X_{t-3} + u_{t-2})$$

$$= \delta_1 X_{t-1} + \beta_1 \delta_1 X_{t-2} + \beta_1^2 \delta_1 X_{t-3} + \dots$$